Reg No.:_____ Name:_____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Third Semester B.Tech Degree Examination December 2020 (2019 Scheme)

Course Code: CHT203
Course Name: CHEMICAL PROCESS PRINCIPLES

Max. Marks: 100 Duration: 3 Hours

Humidity Chart may be Permitted

PART A

Answer all questions. Each question carries 3 marks Marks

1	Calculate Molality and Molarity of a 20% aqueous NaCl solution. The density of the solution is 1.148 g/cm^3.	(3)
2	Calculate the density of CO_2 in kg/m^3 at 525 K and 200 kPa. Assume that CO_2 obeys ideal gas law.	(3)
3	State Henry's law and Raoult's Law. List their applications	(3)
4	Explain the vapour liquid equilibrium of non-ideal solution with neat diagram.	(3)
5	Wet paper containing 75% water is dried to remove 80% of water present. Determine the composition of the dried paper.	(3)
6	A weak acid containing 14.5 % H_2SO_4 and the rest water is fortified by adding 500 kg of concentrated acid containing 80% H_2SO_4. Determine the amount of solution obtained if it contains 19.5% H_2SO_4.	(3)
7	Find the % excess air supplied when 50 kg sulphur is burned with 90 kg O_2.	(3)
8	C and O_2 are mixed in the ratio 3:4. One mole of CO_2 and one mole of CO are formed. Find the conversion and yield of CO_2.	(3)
9	State Hess's law of constant heat summation and explain with an example.	(3)
10	With the help of an example, explain the terms (i) standard heat of reaction, (ii) standard heat of formation and (iii) standard heat of combustion.	(3)

PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

11	a)	A gaseous mixture has the following composition by volume: methane - 10%, ethane - 30% and hydrogen - 60% at 303 K and 2 atm pressure. Calculate: (i) Composition in mol%, (ii) Composition in weight%, (iii) Average Molecular weight and (iv) Density of the mixture.	(10)
	b)	The specific gravity of a hydrocarbon oil is 0.88 at 288.8 K. Determine the corresponding values in Baume and API scales.	(4)

12 a) The effective heat capacity of a mixture of gases is given by: (8)

$$C_p = 7.13 + 0.577 \times 10^{-3}t + 0.0248 \times 10^{-6}t^2$$

where Cp is in Btu/(lb.^{0}F) and t is in ^{0}F. Change the equation into the form in which Cp is given in kJ/kg.K and temperature in K.

b A benzene solution of an organic compound A analyses 10% A. The molality of solution is 0.62. Calculate: (i) The molecular weight of compound A and (ii) the mole fraction of the compound A in the solution. (6)

Module 2

13 The dry bulb temperature and dew point of an air sample at 101.3 kPa are 328 K and 308 K respectively. Using Psychrometric chart, calculate: (i) absolute humidity, (ii) percent saturation, (iii) relative saturation, (iv) Humid volume, (v) humid heat and (vi) wet bulb temperature. (14)

14 a) Carbon dioxide at 300 K and 100 kPa is saturated with water vapour. The vapour pressure of water at 300 K is 3.56 kPa. Determine: (i) the mole percent of water in the mixture, (ii) the percent composition by weight and (iii) the amount of vapour in kg/m^3 of the mixture (9)

b) N-heptane and toluene form an ideal solution. At 373 K their vapour pressures are 106 kPa and 74 kPa respectively. Determine the composition of liquid and the vapour in equilibrium at 373 K and 101.3 kPa. (5)

Module 3

15 a) Sodium carbonate is recovered as decahydrate from 1000 kg of a 5% solution of Na_2CO_3 in water. If it is desired that 95% of Na_2CO_3 is to be recovered as decahydrate by cooling to 278 K, determine the following: (10)

(i) The mass of water evaporated and

(ii) The mass of crystals formed.

The solubility of Na_2CO_3 in water at 278 K is 9% (weight)

b) A batch of soyabean containing 18% oil and 10% water is treated with pure hexane to give a residue containing 0.5% oil and 20% hexane. Find the percentage recovery of oil, if 1000 kg of soyabean was treated. (4)

16 a) A feed mixture containing 40% benzene, 30% toluene and 30% xylene is fed to a distillation column. The distillate contains 99.5% benzene and 0.5% toluene. The residue from this unit is fed to another distillation column from which distillate of composition 97% toluene, 2% benzene and 1% xylene and a residue of composition 5% toluene and 95% xylene are obtained. Calculate the composition of the residue from the first distillation column. (9)

b) A mixture is separated into two fractions. The first fraction contains 50% A and 50 % B. The second fraction contains 25% A and 75 % B. The weight of second fraction is twice that of first. Find the composition of the original mixture. (5)

Module 4

17 a) A Coal sample containing 80% carbon and 6% ash by weight, when burned leaves a cinder which contains 90% ash and 10% carbon. If 100 kg of coal is charged, calculate: (i) the weight of cinder produced and (ii) the percent fuel wasted. (8)

b) Explain recycle and purge with the help of an example. (6)

18 a) Pure sulphur is burned with excess air to give sulphur trioxide. Due to incomplete oxidation some SO_2 is also formed. The analysis of the burner gases showed 8.2% SO_3, 0.91% SO_2, 8.65% O_2 and 89.24 % N_2. Determine: (i) Percent excess air and (ii) Quantity of flue gas per kg of sulphur burned. (8)

b) A gas containing only methane and nitrogen is burned with air yielding a flue gas that has an Orsat analysis of CO_2 - 8.75, CO - 1.0%, O_2 - 2.0% and N_2 - 88.3 %. Calculate: (i) The percent composition of fuel and (ii) Percent excess air used. (6)

Module 5

19 a) Carbon monoxide at 100^0C and atmospheric pressure is burnt with 100% excess dry air at 200^0C. The products of combustion leave the reaction chamber at 800^0C. Calculate the heat evolved in the reaction chamber in kcal/kmol of CO burnt assuming complete combustion. Standard heat of combustion of CO is -67.636 kcal/kmol. Mean heat capacities of CO, air, carbon dioxide, oxygen and nitrogen are 7.02, 7.23, 11.92, 7.94 and 7.06 kcal/kmol respectively. (10)

b) Explain the methods used for estimation of heat of fusion and vaporization. (4)

20 a) A gas mixture consisting of 20% ethane and 80% oxygen at 298 K is burned completely after diluting with double the volume of CO_2, also at 298 K. The mean heat capacities in J/mol K are 54.56 for CO_2, 43.02 for water vapour and 35.52 for oxygen. The standard heat of reaction is -1428 kJ/mol of ethane at 298 K. Determine the theoretical flame temperature. (10)

b) Derive Clausius Clapeyron equation. State its applications. (4)

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